# 8 - Reaction kinetics

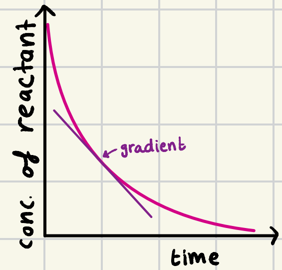
18/12/24

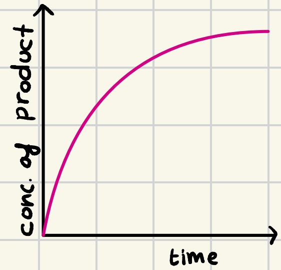
## Rate of reaction

- change in concentration of reactants / products per unit time
- rate of reaction is inversely proportional to time

* rate of reaction is the gradient of curve

* steeper gradient = higher rate of reaction




## Collision theory

- for particles to react with each other, they must collide in the correct orientation with energy greater than or equal to activation energy ($E_A$)

- activation energy is the minimum energy that colliding particles must have for a successful collision to take place [every reaction has a specific $E_A$]

* if the collisions results in the reactants changing into products, it's a successful / effective collision

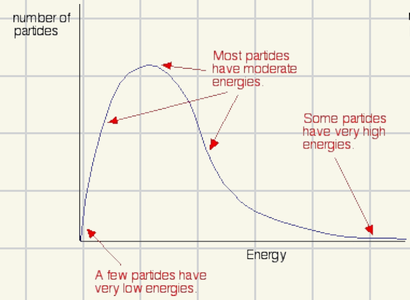
⇒ increase in concentrations or pressure

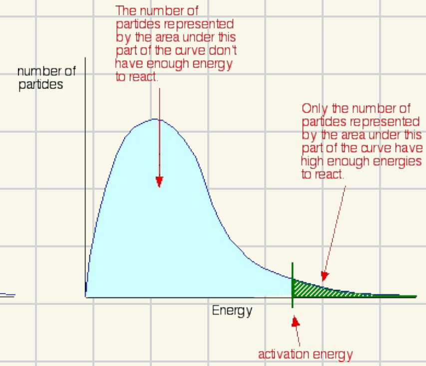
more particles per unit volume, so the particles collide with a greater frequency thus there will be a higher frequency of effective collisions

## Boltzmann distribution

- shows the distribution of energies that molecules of a gas or liquid have at a particular temperature

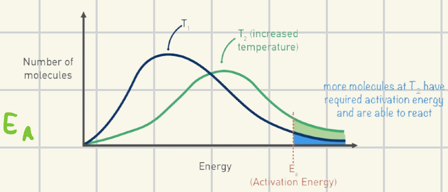
- particles that don't have enough energy can gain energy through collisions




⇒ increase in temperature

particles gain energy, they collide more frequently, thus more particles have energy greater than the $E_A$




⇒ addition of a catalyst

it will lower activation energy, thus particles will be able to have successful / effective collisions